



Assessment

Individual Design Workbook Design Matching Games

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1. Introduction

Modern software engineering frameworks prioritize user-centered design and iterative reflection over rigid plan-based specifications, allowing teams to deliver adaptable digital products that closely align with end-user expectations (Sommerville, 2016). Within the context of interactive educational applications, aligning software architecture with early childhood cognitive capabilities requires structured methodologies that blend Agile project management, design thinking, and human-computer interaction principles (Brown, 2008). This workbook examines the design reasoning, technical spikes, and portfolio contributions developed for an interactive educational module focused on jazz history. By systematically executing iterative prototyping, applying Gestalt visual perception laws, and managing design trade-offs, complex historical data can be converted into intuitive digital media for early learners (Koffka, 1935; Norman, 2013). Furthermore, this report details the integration of empirical feedback obtained during project reviews to justify continuous layout adaptations and narrative refinements (Schwaber & Sutherland, 2020). Through transparent documentation of individual efforts and team collaboration dynamics, this analysis demonstrates how reflective software engineering practices enhance product usability, reduce cognitive overhead, and support progressive learning outcomes in early childhood gamification.

2. Executive Summary & Personal Contribution Identification

In software engineering, modern product development has shifted from rigid, plan-based specifications toward human-centered, iterative methodologies (Sommerville, 2016). Within our team's interactive software project, my primary responsibility was leading the design, conceptualization, and user experience (UX) architecture for an educational Jazz Musician and Instrument Matching Game. This module aims to introduce young learners to rich cultural music history through gamified learning.

Primary contributions to the team design portfolio focused on domain research, asset elicitation, and visual adaptation tailored specifically to early childhood educational standards. To establish a rigorous pedagogical foundation, extensive scholarly and historical research was conducted to source authentic visual references for twelve seminal jazz figures, including Art Blakey, John Coltrane, Miles Davis, Thelonious Monk, Benny Goodman, Billie Holiday, Django Reinhardt, Glenn Miller, Charles

Mingus, Lionel Hampton, Stéphane Grappelli, and Toots Thielemans. Recognizing that complex historical photographs could impose an excessive cognitive load on young learners, a systematic visual transformation process was executed to reconceptualize these archival images into vibrant cartoon illustrations designed to align with the developmental and cognitive capacities of early childhood users.

Beyond visual asset synthesis, core responsibilities encompassed technical exploration, narrative design, and interactive prototyping. Exploratory technical spikes were performed on the KOPO platform to systematically analyze its architectural constraints, spatial boundaries, and underlying matching game interaction mechanics. To complement these functional mechanics, localized child-friendly narrative backgrounds were formulated, thematic visual environments were curated, and playful auditory cues, such as cheerful background music, were incorporated to foster emotional engagement and positive reinforcement. Ultimately, these functional and aesthetic components were synthesized into clickable interactive prototypes across low, medium, and high fidelities within Figma, establishing intuitive navigation architecture, structured spatial layouts, and optimized user experience flows.

3. Design Reasoning and Decision-Making Rationale

3.1 Applying Design Thinking and Human-Centered UX

The development of the matching game relies heavily on the design thinking framework: divergent thinking involves the continuous generation of ideas and the exploration of gameplay mechanics, while convergent thinking entails the ongoing selection and refinement of elements to ensure a smooth flow within the game (Brown, 2008).

During the initial Empathise and Define stages, the target user persona was established as six-year-old early learners. Children at this developmental milestone rely primarily on visual recognition, immediate sensory feedback, and intuitive spatial associations rather than dense textual descriptions (Piaget, 1962). To maximize cognitive accessibility and minimize visual noise (Garner & Alley, 2013), the user interface was structured using key Gestalt Principles (Koffka, 1935). Specifically, the principle of proximity was applied by grouping musician avatars directly adjacent to their target instrument drop-zones to minimize extraneous cognitive load. Furthermore, the principle of similarity was enforced by maintaining consistent visual styling, color palettes, and cartoon aesthetics across all twelve musician cards so users intuitively understand that these cards share identical interactive behaviors.

Extending beyond initial interface structuring, the design process encompassed technical exploration, narrative design, and multi-fidelity interactive prototyping. Exploratory technical spikes were conducted on the KOPO platform to systematically

evaluate architectural constraints, spatial boundaries, and core matching game interaction mechanics. To complement these functional mechanics, localized child-friendly narrative backstories were formulated, thematic visual environments were curated, and cheerful auditory feedback was integrated to cultivate emotional engagement. Ultimately, these functional and aesthetic components were synthesized into clickable interactive prototypes across low, medium, and high fidelities within Figma, establishing intuitive navigation architecture, structured spatial layouts, and optimized user experience flows.

3.2 Layout Selection and Prototyping Fidelities

To model the user interaction, technical spikes were conducted as focused exploratory experiments aimed at evaluating platform constraints and technical boundaries (Schwaber & Sutherland, 2020). Through benchmarking analysis of the KOPO matching platform, a hybrid grid layout combined with a drill-down architecture was identified as the most effective structure for young users. The grid layout provides structured spatial alignment for visual elements, while the drill-down framework facilitates clear, step-by-step navigation without overencumbering smaller display interfaces.

In alignment with software engineering prototyping practices, the design process progressed through multiple iterative fidelity stages. Low-fidelity paper wireframes were initially developed to rapidly map out basic card placements and structural layouts. Subsequently, medium-fidelity clickable wireframes were constructed to establish navigation flows, touch-target sizing, and spatial arrangements using placeholder elements. Finally, high-fidelity interactive mockups were produced within Figma, fully incorporating color theory principles, custom vector visual assets, and playful auditory cues to simulate the comprehensive end-product user experience.

4. Reflection on Feedback, Iteration, and Constraints

4.1 Empirical Feedback and Stakeholder Iteration

Agile development relies on Empiricism—the principle that knowledge stems from experience and decisions are based on observed feedback through transparency, inspection, and adaptation (Schwaber & Sutherland, 2020). During the project review phase, the course instructor served as the primary stakeholder, providing evaluative feedback on the prototype. The principal critique emphasized that although the initial visual asset selection incorporating authentic historical photographs of twelve jazz musicians maintained historical integrity, it presented an excessively abstract and dense aesthetic that could potentially impose a high cognitive barrier for six-year-old learners.

In response to stakeholder feedback, a comprehensive iterative refinement cycle was

executed to systematically transform the application across visual, auditory, and narrative dimensions. Visually, the initial framework was reframed by vectorizing realistic portraits into bright, approachable cartoon characters, ensuring that all visual assets directly aligned with early childhood cognitive capabilities. From an auditory perspective, cheerful background music and playful reward sounds were integrated upon successful matches to enhance emotional resonance and reinforce positive user behavior. Narratively, dense biographical descriptions were adapted into simplified character story vignettes, such as introducing Miles the Friendly Trumpeter, thereby aligning with the reading comprehension levels of early learners while preserving the core educational value of the product.

4.2 Design Trade-offs and Process Anti-Patterns

Managing project constraints required navigating distinct design trade-offs between historical authenticity and child usability. Preserving exact historical realism was traded off in favor of cartoon abstraction to increase user engagement and acceptability (Sommerville, 2016).

Furthermore, a significant trade-off emerged between project scope and delivery time, as sourcing twelve distinct musicians presented a considerable development challenge. Rather than reducing the historical diversity of the content, modular component design in Figma was leveraged to streamline visual asset production and maintain the planned scope within established timelines.

Reflecting on the team sprint retrospective, minor workflow anti-patterns were identified and corrected. Initially, a risk existed of treating the sprint review as a static theatrical performance rather than a collaborative learning checkpoint (Kniberg, 2011). Recognizing this limitation, team focus shifted toward fostering transparent discussions around design uncertainty, utilizing Figma artifacts as concrete evidence for design reasoning.

5. Conclusion

Bringing the Jazz Musician Matching Game to life was a vibrant journey that transformed rigid engineering frameworks into a playful reality. We wove together Design Thinking, Gestalt principles, direct stakeholder feedback, and rapid Figma prototyping to distill complex historical archives into an accessible and engaging tool for children. This project beautifully proves that systematic design, smart compromises, and relentless reflection are the true cornerstones of building software that genuinely puts the user first.

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